

SIMATS
ONLINE
LAB
ENGINEERING

SIMATS Online Programming Lab

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Array Rotation

ARRAY ROTATION

"Implement array rotation (left/right) for arrays of size 10 by K positions without using extra space. Use a reversal algorithm. The program must validate K ($0 \leq K < N$), show original→rotated→reversal steps visually, compute the sum difference, and support multiple rotations.

Sample Input/Output:

Enter 10 numbers: 1 2 3 4 5 6 7 8 9 10

Rotate Left (L)/Right (R): L

By positions: 3

Original: [1 2 3 4 5 6 7 8 9 10]

Step1: Reverse 0-2 → [3 2 1 4 5 6 7 8 9 10]

Step2: Reverse 3-9 → [3 2 1 10 9 8 7 6 5 4]

Step3: Reverse all → [4 5 6 7 8 9 10 1 2 3]

Sum difference: 0"

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 void reverse(int arr[], int start, int end)
3 {
4     while(start<end)
5     {
6         int temp = arr[start];
7         arr[start] = arr[end];
8         arr[end] = temp;
9         start++;
10        end--;
11    }
12 }
13 void printArray(int arr[], int n)
14 {
15     int i;
16     for(i=0; i<n ; i++)
17         print("%d", arr[i]);
18     print("\n");
19 }
20 int main()
21 {
```

OUTPUT

Your OUTPUT will appear here!

INPUT

1 2 3 4 5 6 7 8 9 10

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